

Predicting Hospital Length of Stay

GPH-GU 2353: Linear Regression and Modelling

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Figure 1: Univariate Analysis of Categorical Variables

	Overall (N=100000)
Gender	
Male	42357 (42.4%)
Female	57643 (57.6%)
End Stage Renal Disease	
No	96358 (96.4%)
Yes	3642 (3.6%)
Asthma	
No	96473 (96.5%)
Yes	3527 (3.5%)
Iron Deficiency	
No	90506 (90.5%)
Yes	9494 (9.5%)
Pneumonia	
No	96055 (96.1%)
Yes	3945 (3.9%)
Substance Dependence	
No	93694 (93.7%)
Yes	6306 (6.3%)
Major Psychological Disorder	
No	76096 (76.1%)
Yes	23904 (23.9%)
Fibrosis	
No	99521 (99.5%)
Yes	479 (0.5%)
Malnutrition	
No	95052 (95.1%)
Yes	4948 (4.9%)
Blood Disease	
No	92000 (92.0%)
Yes	8000 (8.0%)

Figure 2: Univariate Analysis of Continuous Variables

	Overall (N=100000)
Hematocrit Level (g/dL)	
Mean (SD)	12.0 (2.03)
Median [Min, Max]	11.9 [4.40, 24.1]
Neutrophils Level (cells/microL)	
Mean (SD)	10.2 (5.35)
Median [Min, Max]	9.40 [0.100, 246]
Sodium Level (mmol/L)	
Mean (SD)	138 (3.00)
Median [Min, Max]	138 [125, 151]
Glucose Level (mmol/L)	
Mean (SD)	142 (30.0)
Median [Min, Max]	142 [-1.01, 271]
Blood Urea Nitrogen Level (mg/dL)	
Mean (SD)	14.1 (13.0)
Median [Min, Max]	12.0 [1.00, 683]
Creatinine Level (mg/dL)	
Mean (SD)	1.10 (0.200)
Median [Min, Max]	1.10 [0.220, 2.04]
BMI	
Mean (SD)	29.8 (2.00)
Median [Min, Max]	29.8 [22.0, 38.9]
Pulse (Beats/Minute)	
Mean (SD)	73.4 (11.6)
Median [Min, Max]	73.0 [21.0, 130]
Respiration (Breaths/Minute)	
Mean (SD)	6.49 (0.568)
Median [Min, Max]	6.50 [0.200, 10.0]
Length of Stay (Days)	
Mean (SD)	4.00 (2.36)
Median [Min, Max]	4.00 [1.00, 17.0]

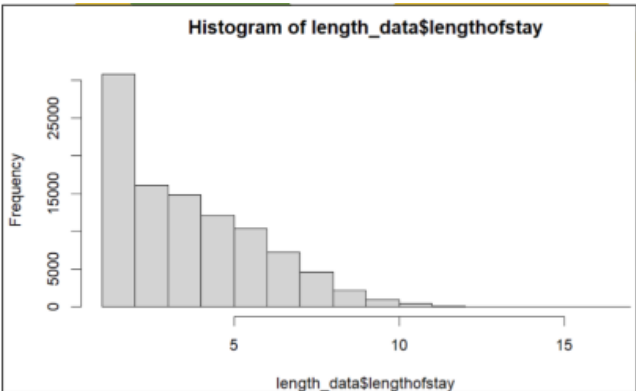
Introduction

The goal of this study is to develop a reliable method for estimating how long patients will stay in the hospital. This is crucial for planning patient care and managing hospital resources effectively. Our aim is to develop a model that best predicts this duration. The dataset used for this analysis was accessed from [Kaggle](#). It is a large dataset containing visit-level information on 100,000 patients admitted to hospitals. These data include various details on patient health and how long they stayed in the hospital. Our main outcome is the variable "length of stay" (LOS): the number of days a patient spends in the hospital. Additional variables used in the regression analysis include disease status (such as asthma, iron deficiency, and other), levels of different health biomarkers (such as hematocrit, glucose, etc.), and additional information such as how many times the patient has been readmitted within the last six months.

Upon initial review of the dataset, we removed variables that were irrelevant to the analysis or that did not provide sufficient detail for interpretation. These variables included the patient and hospital facility identifiers, among other miscellaneous variables.

Figure 1 displays the univariate analysis of the categorical variables in the dataset; Figure 2 displays the univariate analysis of continuous variables in the dataset. All categorical variables were converted to factors prior to analysis. Missing values were checked; no missing values were present in the dataset. Below, Figure 3 depicts a histogram of length of stay, the outcome variable of interest.

Figure 3: Histogram of Length of Stay



Unadjusted Model

The unadjusted model used hospital length of stay as the outcome variable, with all remaining variables in the dataset included as predictors. In the unadjusted model, the blood disease status variable (variable = "hemo1") had the largest Beta value; an individual with a blood disease could expect to stay in the hospital for an additional 1.32 days on average, compared with an individual without a blood disease, after adjusting for all other variables in the model.

The unadjusted model had an R-squared value of 0.18, indicating limited predictive power. Many predictors, particularly the continuous health biomarker variables, were found to be insignificant, indicating their minimal impact on length of stay prediction. Further refinement is needed to enhance the model's accuracy and relevance.

Figure 4: R Output Summary of Unadjusted Model

```
Call:
lm(formula = lengthofstay ~ ., data = length)

Residuals:
    Min       1Q   Median       3Q      Max
-7.1446 -1.6080 -0.4972  1.4822 12.3219

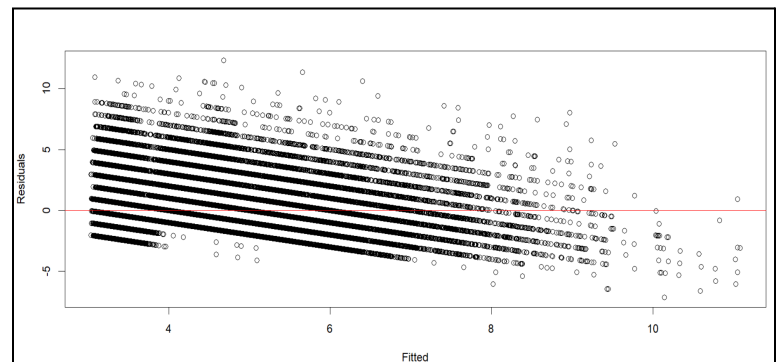
Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  3.2786906   0.3442518   9.524 < 2e-16 ***
genderM      0.0486040   0.0146774   3.311 0.000928 ***
dialysisrenalendstage1  0.9006437   0.0386833  23.283 < 2e-16 ***
asthma1     0.8795349   0.0368656  23.858 < 2e-16 ***
irondef1    0.7395530   0.0247548  29.875 < 2e-16 ***
pneum1     0.6656323   0.0357604  18.614 < 2e-16 ***
substancedependence1  0.9348700   0.0283334  32.995 < 2e-16 ***
psychologicaldisordermajor1 1.3069526   0.0164382  79.507 < 2e-16 ***
fibrosisandother1  1.1354976   0.1001377  11.339 < 2e-16 ***
malnutrition1  0.6203158   0.0342167  18.129 < 2e-16 ***
hemo1      1.3219707   0.0268892  49.164 < 2e-16 ***
hematocrit  0.0091700   0.0037840   2.423 0.015381 *
neutrophils 0.0180604   0.0013250  13.630 < 2e-16 ***
sodium     -0.0024444   0.0022560   -1.084 0.278582
glucose    -0.0001602   0.0002256   -0.710 0.477801
bloodureanitro 0.0102153   0.0005619   18.181 < 2e-16 ***
creatinine -0.0297519   0.0337933   -0.880 0.378641
bmi        0.0016891   0.0033773    0.500 0.616985
pulse      0.0007163   0.0005812    1.232 0.217767
respiration -0.0184611   0.0122635   -1.505 0.132234
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.14 on 99980 degrees of freedom
Multiple R-squared:  0.1783,    Adjusted R-squared:  0.1781
F-statistic: 1142 on 19 and 99980 DF, p-value: < 2.2e-16
```

Testing Assumptions: Issues in the Model

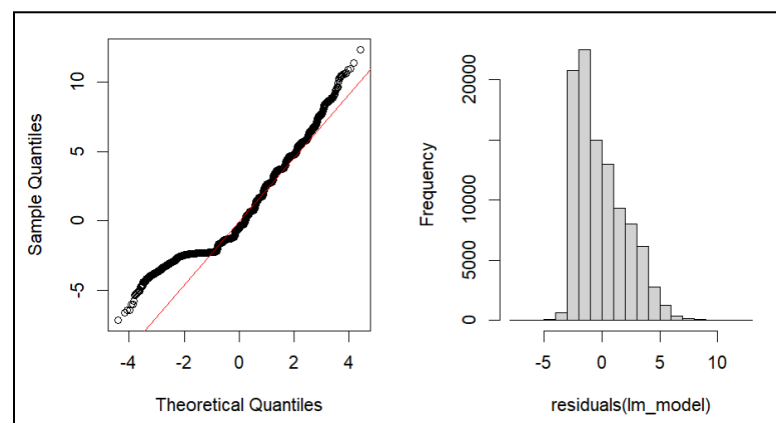
Figure 5 displays the Residuals vs. Fitted plot for the Unadjusted Model. The plot displays possible non-linearity, as evidenced by the generally negatively sloped spread of residuals along the fitted line. The plot also indicates the presence of potential outliers at the upper-end positive values on the x-axis. Generally, the plot does not indicate heteroscedasticity.

Figure 5: Residuals vs. Fitted Plot - Unadjusted Model



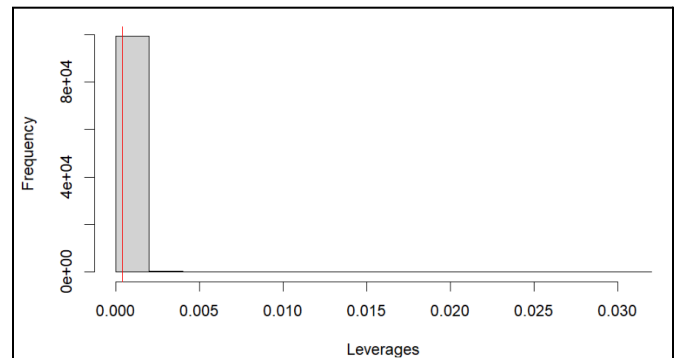
Normality of errors was checked using a Q-Q plot and histogram, shown in Figure 6. The plots indicate potential non-normality, evidenced by the slight "S"-type pattern in the Q-Q plot and the positive skew of the histogram. As the plots do not indicate extreme non-normality of the errors, we did not adjust for normality when performing model selection.

Figure 6: Q-Q Plot and Histogram of Residuals - Unadjusted Model



The presence of outliers was indicated by the Residuals vs. Fitted Plot in Figure 5. The presence of large leverage values was assessed for by calculating hat values for the Unadjusted Model; any value with a leverage twice as large as the average hat value was considered to be “large.” The mean hat value was 0.0002; an analysis of all leverage values twice as large as the average value identified a significant number of large leverage values in the dataset. Figure 8 shows a histogram of the leverage values, where the red line represents the hat value cut-off. As the majority of values in the dataset had large leverages, we did not adjust for leverage values in our model selection. We next assessed for influential values using Cook’s distances. No Cook’s distances above the 0.5 threshold (indicating a “large” distance) were present in the dataset.

Figure 7: Histogram of Leverage Values - Unadjusted Model



We assessed for multicollinearity by calculating the Variance Inflation Factor (VIF) in the unadjusted model; all predictors had VIF values of about 2 or lower. As no values were above 10, it was assumed that no multicollinearity is present in the dataset.

Model Adjustment

Our diagnostic tests indicated issues with non-linearity and potential outliers to be addressed in our final model.

To address the issue of non-linearity, we employed the following methods: 1) Performing a box-cox power transformation of the outcome variable; 2) Using log transformations of the outcome variable; 3) Checking for serial correlation; and 4) Transforming predictors using polynomials.

The polynomial method was the most successful. Figure 8 shows that each continuous variable in the model was taken to its third-order value to assess for significance. We removed the highest-order non-significant variables, as applicable, and added the remaining variables to a polynomial-adjusted model. Figure 9 shows the Residuals vs. Fitted plot for the polynomial-adjusted model; the issue of non-linearity appears to be slightly addressed, as less of discernable pattern is noticeable along the spread of residuals along the fitted line.

```
summary(lm(lengthofstay ~ gender +
+ dialysisrenalendstage
+ asthma
+ irondef
+ pneum
+ substancedependence
+ psychologicaldisordermajor
+ fibrosisandother
+ malnutrition
+ hemo
+ hematocrit + I(hematocrit^2) + I(hematocrit*hematocrit^2)
+ neutrophils + I(neutrophils^2) + I(neutrophils*neutrophils^2)
+ sodium + I(sodium^2) + I(sodium*sodium^2)
+ glucose + I(glucose^2) + I(glucose*glucose^2)
+ bloodureanitro + I(bloodureanitro^2) + I(bloodureanitro*bloodureanitro^2)
+ creatinine + I(creatinine^2) + I(creatinine*creatinine^2)
+ bmi + I(bmi^2) + I(bmi*bmi^2)
+ pulse + I(pulse^2) + I(pulse*pulse^2)
+ respiration + I(respiration^2) + I(respiration*respiration^2),
length_scaled))
```

Figure 8: Adding Polynomials to Model

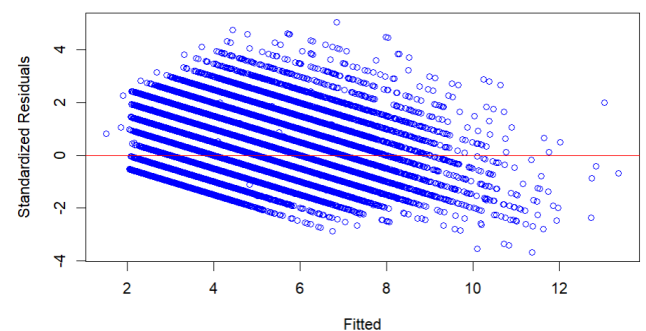


Figure 9: Residuals vs. Fitted Plot - Polynomial-Adjusted Model

To confirm the presence of suspected outliers, we calculated studentized residuals and the Bonferroni Critical Value based on the t-distribution. A total of 286 data points with residuals higher than the critical value of 3.6 were identified. A dataset was built that excluded each of these 286 points, and an unadjusted regression was run (outcome variable = length of stay; predictor variables = all remaining variables in the dataset) to compare with the original Unadjusted Model. Figure 10 shows the output of the unadjusted model using the dataset with the 286 outlier points removed; as the adjusted R-squared value is the same as the original model (0.18), the outliers will not be removed from the dataset in the Final Model.

Figure 10: Summary Output of Unadjusted Model using Dataset with Outliers Removed

```

Coefficients:
(Intercept)      3.306288    0.010033 329.554 < 2e-16 ***
genderM          0.048741    0.014670   3.323 0.000892 ***
dialysisrenalendstage1 0.898746    0.038667  23.243 < 2e-16 ***
asthma1         0.880178    0.036846  23.888 < 2e-16 ***
irondef1        0.740473    0.024742  29.928 < 2e-16 ***
pneum1          0.663269    0.035746  18.555 < 2e-16 ***
substancedependence1 0.935864    0.028318  33.048 < 2e-16 ***
psychologicaldisordermajor1 1.307666    0.016429  79.593 < 2e-16 ***
fibrosisandother1 1.135917    0.100083  11.350 < 2e-16 ***
malnutrition1    0.621810    0.034198  18.182 < 2e-16 ***
hemo1           1.319886    0.026877  49.109 < 2e-16 ***
hematocrit       0.019094    0.007685   2.485 0.012965 *
neutrophils      0.097003    0.007089  13.683 < 2e-16 ***
sodium          -0.007350    0.006764  -1.087 0.277154
glucose         -0.004427    0.006764  -0.655 0.512758
bloodureanitro  0.131992    0.007274  18.146 < 2e-16 ***
creatinine      -0.006280    0.006764  -0.928 0.353191
bmi              0.003216    0.006764   0.475 0.634471
pulse           0.008108    0.006764   1.199 0.230657
respiration     -0.010571    0.006968  -1.517 0.129269
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

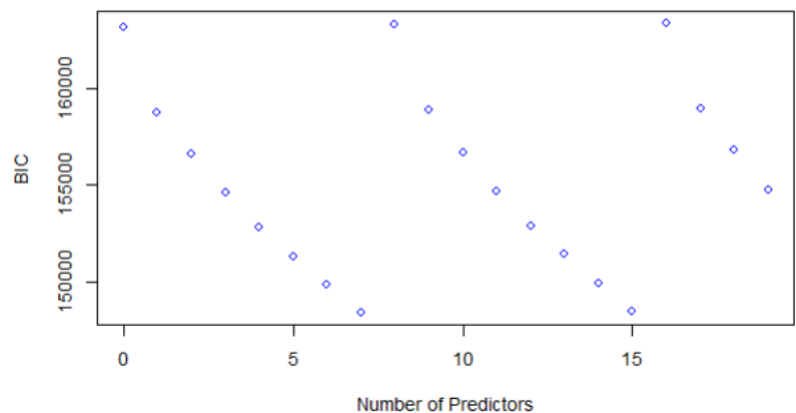
Residual standard error: 2.139 on 99976 degrees of freedom
Multiple R-squared:  0.1784,    Adjusted R-squared:  0.1782
F-statistic: 1142 on 19 and 99976 DF,  p-value: < 2.2e-16

```

Model Selection

Our Adjusted Model included the significant polynomials added to the model during the adjustment procedures. We calculated Bayesian information criterion (BIC) using this Adjusted Model to select a Final Model. Figure 11 demonstrates that the 8-predictor model has the lowest BIC value (148384.5).

Figure 11: Plot of BIC vs Number of Predictors



Final Model

Our final model incorporates the BIC-selected adjustments made during the diagnostic model selection processes; the 9th-20th variables in our Unadjusted Model were removed as the 8-predictor model was selected in the BIC process.

Figure 12 shows the summary output of the Final Model; note that the adjusted R-Squared Value has decreased (0.1446), compared with the original model (0.18).

Figure 12: Summary Output of Final Model

```

Call:
lm(formula = lengthofstay ~ gender + +dialysisrenalendstage +
    asthma + irondef + pneum + substancedependence + psychologicaldisordermajor,
    data = length_scaled)

Residuals:
    Min       1Q   Median       3Q      Max
-5.0866 -1.7324 -0.4232  1.5768 13.6139

Coefficients:
(Intercept)      3.38610    0.00989 342.387 < 2e-16 ***
genderM          0.03711    0.01424   2.606 0.00915 **
dialysisrenalendstage1 1.45939    0.03762  38.789 < 2e-16 ***
asthma1         0.90662    0.03759  24.116 < 2e-16 ***
irondef1        0.99184    0.02431  40.795 < 2e-16 ***
pneum1          0.82847    0.03626  22.846 < 2e-16 ***
substancedependence1 1.03733    0.02865  36.212 < 2e-16 ***
psychologicaldisordermajor1 1.34630    0.01657  81.236 < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

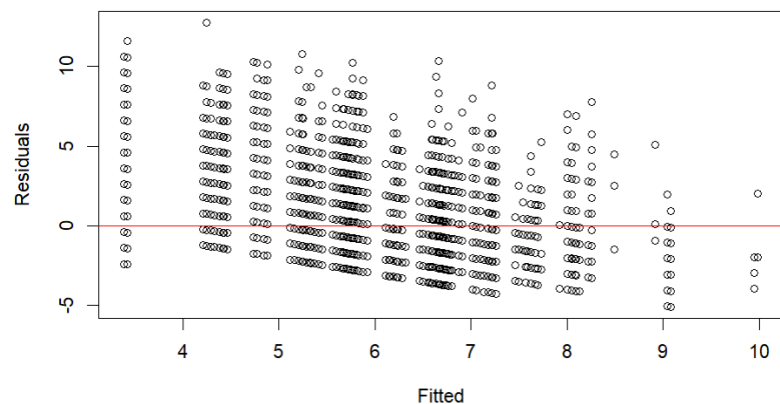
Residual standard error: 2.183 on 99992 degrees of freedom
Multiple R-squared:  0.1447,    Adjusted R-squared:  0.1446
F-statistic: 2416 on 7 and 99992 DF,  p-value: < 2.2e-16

```


The Final Model does not include any of the continuous health biomarkers variables, nor any of the polynomials added during the Model Adjustment; rather, only categorical variables are included as predictors. The end-stage renal disease status variable has the largest Beta value; individuals with end stage renal disease would expect to stay in the hospital for an additional 1.46 days on average, compared with individuals without end-stage renal disease, while controlling for all other variables in the model. Interestingly, a Male in the Final Model would be expected to have a 0.04 day longer hospital stay, compared with a Female, after adjusting for all other predictors in the model. The presence of a psychological disorder was also found to be a highly significant predictor of length of hospital stay; an individual with a psychological disorder would expect to stay in the hospital for an additional 1.35 days, compared with someone without a psychological disorder, after controlling for all other variables in the model.

Figure 13: Residuals vs. Fitted Plot - Final Model

Figure 13 shows the Residual vs. Fitted Plot of the Final Model; it shows that the issue of non-linearity was not addressed in the final model.



An analysis of variance (ANOVA) test was conducted to compare the Final Model with the original Unadjusted Model; the null hypothesis being that both models are equal. A significant p-value was found; the null hypothesis was rejected, and the Final Model is determined to be significantly different from the original Unadjusted Model.

Summary

Utilizing data from a robust sample of 100,000 patients, this study delves into the influence of various health indicators and conditions on hospitalization duration. While the Unadjusted Model underscores the significance of factors like disease/condition status and health indicators, it faces challenges such as limited predictive power, heteroskedasticity, and insignificance of certain predictors. To mitigate these issues, diagnostic testing was undertaken. Polynomial transformations were implemented and outlier removal was considered but ultimately not undertaken. The model selection process, using BIC, resulted in the removal of our continuous health biomarker predictors, in addition to the additional polynomial-adjusted predictors, from the Final Model. Despite a decreased adjusted R-squared value in the Final Model, the identification of end-stage renal disease as a key predictor of length of stay underscores its clinical relevance.

Moving forward, future research may include examining the predictive power of specific conditions and exploring hospital bed space modeling. Addressing limitations like heteroskedasticity and large leverages is paramount for refining model accuracy. Additionally, considering length of stay as a categorical variable could provide a more nuanced analysis. In summary, this study contributes valuable insights into predicting hospital length of stay and suggests avenues for further research and model refinement.